Section 1: Case Prompt

GreenLoop Packaging is a mid-sized manufacturer of compostable food containers based in Pune, India. Over the last three years its revenue has grown by 18% annually, but gross margin has fallen from 34% to 27% because the price of its main input, sugarcane bagasse, has risen while large restaurant-chain customers have demanded lower prices. The CEO is deciding between three options: (a) build a second plant closer to the sugarcane belt, (b) sign a long-term supply contract with a single bagasse supplier, or (c) develop a premium product line for health-conscious retail consumers. Instructions: Your submission must (1) identify the core problem, (2) evaluate each option against cost, risk and speed, (3) recommend one option with a clear implementation plan, and (4) list the assumptions on which your recommendation depends. Do not exceed 1,000 words. Cite any outside data you use.

Section 2: Analysis

The root problem is not demand; it is that GreenLoop buys its main input at a price it cannot control and sells to customers who can. Revenue is up 18% a year yet margin has dropped seven points, so every extra tray sold earns less than the last. Option (a), a second plant near the cane belt, attacks the cost directly. Bagasse is bulky and cheap per kilo, so trucking it 600 km from the sugar mills to Pune eats a large share of its value; a plant next to the mills could plausibly cut landed input cost by 12 to 15 percent in my estimate. The downside is capital: I assume roughly 90 crore rupees and an 18-month build. Option (b) leaves GreenLoop dependent on one supplier, and (c) does nothing about the cost base. I therefore recommend (a), phased: lease a shed near Kolhapur in the first six months to prove the economics before committing to a full build. Key assumptions: mill owners will give long-term access to fresh bagasse, and the restaurant chains will not demand further price cuts during construction.